Power Budget Examp

Team Number:	204
Project Name:	**Heart Rate Monitor**
Team Member Names:	**Jaxon, Alessandro, Connor, Christian**
Version:	**1**

A. List ALL major components (active devices, integrated circuits, etc.) except for pow

All Major Components	Component Name	Part Number	yVoltageRan
	Microcontroller Board	PIC18F57Q43 Curiosity Nano	*+5
	DAC	MCP4921-E/P	*+5
	Audio Amplifier	LM386N-3	*+5
		CMS-23558N-L152	
	Speaker		*+5
	Green LED (debug)	standard LED	*+5
	Red LED (alarm)	standard LED	*+5

B. Assign each major component above to ONE power rail below. Try to minimize the r

+5V Power Rail	Component Name	Part Number	yVoltageRan
	Microcontroller Board	PIC18F57Q43 Curiosity Nano	*+5
	DAC	MCP4921-E/P	*+5
	Audio Amplifier	LM386N-3	*+5
	Green LED (debug)	standard LED	*+5
	Red LED (alarm)	standard LED	*+5
			Total
c1. Regulator or Source Ch	+5V Regulator	LM7805	*+7 to *+35
			Total Remaining

C. For each power rail above, select a specific voltage regulator using the same proces

D. Select a specific external power source (wall supply or battery) for your system, and

External Power Source 1	Component Name	Part Number	yVoltageRan
Power Source 1 Selection	Plug-in Wall Supply	9V DC Adapter	+120 VAC
Power Rails Connected to	+5V Regulator	LM7805	+5v to +35v
			Total Remaining Current Availabl

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regu
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a s

ıle

er sources, voltage regulators, resistors, capacitors,			
#	**ıteMaximumCurrent (**	**alCurrent(mA)**	**Unit**
1	50	50	mA
1	1	1	mA
1	100	100	mA
	included with audio		
1	amplifier	0	mA
1	10	10	mA
1	10	350	mA
ıumber of different power rails in the design.			
#	**ıteMaximumCurrent (**	**alCurrent(mA)**	**Unit**
1	50	50	mA
1	1	1	mA
1	100	100	mA
1	10	10	mA
1	10	10	mA
	Subtotal	171	mA
	Safety Margin	25%	
Current Required on +5V Rail		213.75	mA
1	1000	1000	mA
Current Available on +5V Rail		786.25	mA
ss as for major component selection. Confirm that			
l confirm that it can supply all of the regulators for			
Output	**ıteMaximumCurrent (**	**alCurrent(mA)**	**Unit**
+9 VAD	1000	1000	mA
5	1000	214	mA
'e on External Power Source 1		786	mA

ılator (e.g., +14V for a +12V regulator).
eparate power budget for each unit